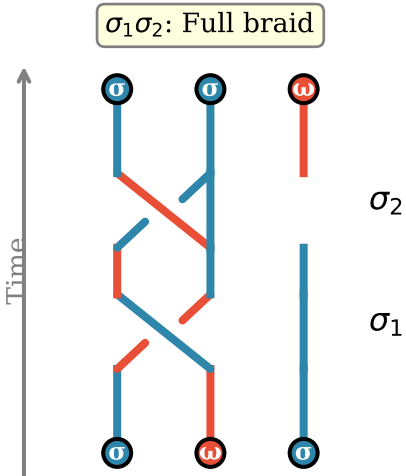


Figure 6: Neglecton Braiding and Universal Gate Operations

(a) Anyon Braiding Worldlines



(b) Braiding Gate Truth Table

Input	Braid	Output	Phase
$ 0\rangle$	σ_1	$ 0\rangle$	1
$ 1\rangle$	σ_1	$ 1\rangle$	$e^{i\pi/4}$
$ +\rangle$	$\sigma_1\sigma_2$	$ -\rangle$	$e^{i\pi/2}$
$ 0\rangle 0\rangle$	σ_1^2	$ 0\rangle 0\rangle$	$e^{i\pi/2}$
$ 1\rangle 1\rangle$	σ_2^2	$ 1\rangle 1\rangle$	-1

(c) R-Matrix: ($\mathfrak{sl}(2)$, $k=2$)

Row object	V_0	V_1	V_2	ω
V_0	1	1	1	1
V_1	1	$e^{i\pi/4}$	$e^{i\pi/2}$	$e^{i\pi/4}$
V_2	1	$e^{i\pi/2}$	-1	$e^{i\pi/2}$
ω	1	$e^{i\pi/4}$	$e^{i\pi/2}$	0
Column object	V_0	V_1	V_2	ω

(d) Universality Gate Overhead

